Lab1-2 Code

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from itertools import product
from typing import Hashable, Tuple, Set, Iterable
import random

Pair=Tuple[Hashable, Hashable]

def reflexive_transitive_closure(A:Iterable[Hashable], R:Set[Pair])->Set[Pair]:
    A=tuple(A)
    n=len(A)
    R=set(R)
    Rstar:Set[Pair]=set()

    for a in A:
        Rstar.add((a,a))

    for i in range(2,n+1):
        for seq in product(A, repeat=i):
            flag=True
            for j in range(i-1):
                if(seq[j],seq[j+1]) not in R:
                    flag=False
                    break
            if flag:
                Rstar.add((seq[0],seq[-1]))

    return Rstar

def rand_R(n:int, edge_prob:float, seed:int=42, include_self_in_R:bool=False):
    rand=random.Random(seed)
    A=list(range(n))
    R:Set[Pair]=set()
    for x in A:
        for y in A:
            if (x==y) and not include_self_in_R:
                continue
            if rand.random()<edge_prob:
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                R.add((x,y))
    return A,R

if __name__=="__main__":
    n=3
    p=0.5

    A, R = rand_R(n, p)

    print("A =", A)
    print("R (random edges) =", sorted(R))

    Rstar= reflexive_transitive_closure(A, R)

    print("\nR* by enumeration =", sorted(Rstar))
```

Output

```
[Running] python -u "/Users/profighted/Desktop/大四上课程PPT/计算与复杂性/作业/Lab1/lab1-2.py"
A = [0, 1, 2]
R (random edges) = [(0, 2), (1, 0), (1, 2)]

R* by enumeration = [(0, 0), (0, 2), (1, 0), (1, 1), (1, 2), (2, 2)]
```

```
[Running] python -u "/Users/profighted/Desktop/大四上课程PPT/计算与复杂性/作业/Lab1/lab1-2.py"
A = [0, 1, 2, 3, 4]
R (random edges) = [(0, 2), (0, 3), (0, 4), (1, 4), (2, 1), (2, 3), (3, 0), (3, 1), (4, 0), (4, 3)]

R* by enumeration = [(0, 0), (0, 1), (0, 2), (0, 3), (0, 4), (1, 0), (1, 1), (1, 2), (1, 3), (1, 4), (2, 0), (2, 1), (2, 2), (2, 3), (2, 4),
(3, 0), (3, 1), (3, 2), (3, 3), (3, 4), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4)]
```

```
[Running] python -u "/Users/profighted/Desktop/大四上课程PPT/计算与复杂性/作业/Lab1/lab1-2.py"
A = [0, 1, 2, 3, 4, 5, 6, 7]
R (random edges) = [(0, 2), (1, 0), (1, 3), (1, 6), (2, 6), (3, 6), (3, 7), (5, 7), (6, 2)]

R* by enumeration = [(0, 0), (0, 2), (0, 6), (1, 0), (1, 1), (1, 2), (1, 3), (1, 6), (1, 7), (2, 2), (2, 6), (3, 2), (3, 3), (3, 6), (3, 7),
(4, 4), (5, 5), (5, 7), (6, 2), (6, 6), (7, 7)]
```